

SAI AMAN GOPISETTI

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PROFESSIONAL SUMMARY

Granted U.S. patent holder and founding R&D engineer who built a graphene-based anode platform from first principles as the sole technical architect across material synthesis, electrode fabrication, electrolyte formulation, coin cell assembly, and full electrochemical validation. Operated without institutional supervision or a technical team for 4+ years, producing independently conceived and experimentally validated results that outperform conventional Li-ion coin cells in capacity, energy density, and operating voltage. Named sole inventor of record on the technical work underlying U.S. Patent No. 12,308,433. Deep hands-on expertise in carbon-based anode chemistry, SEI engineering, ionic liquid electrolyte systems, and electrochemical aging analysis. Experienced in coordinating multi-institutional R&D across sponsored research collaborations and contract testing partnerships.

88.56 mAh First-cycle capacity	537.9 Wh/kg Energy density	4.3 V Operating voltage	68–169% Improvement vs. Li-ion
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INTELLECTUAL PROPERTY

U.S. Patent No. 12,308,433 | “Electrode Materials, Electrodes, Devices, and Methods of Making Thereof” (Granted 2025)

Sole technical inventor. Covers nitrogen and sulfur co-doped porous graphene electrode material composition, solution-based fabrication methodology, and integrated cell architecture for energy storage applications. Conceived, designed, fabricated, and electrochemically validated entirely as the sole technical contributor.

- Experimentally demonstrated first-cycle coin cell capacity of 88.56 mAh, outperforming state-of-the-art carbon-based coin cells (~30 mAh typical), with validated energy density of 537.9 Wh/kg and operating voltage of 4.3V vs. typical 3.6V Li-ion, representing a 68-169% energy density improvement over conventional technology.
- Executed full electrochemical aging protocol across 32 charge/discharge cycles with rigorous galvanostatic cycling and CV-based tracking of coulombic efficiency (CE), capacity retention, capacity fade, impedance growth, and state of health (SoH).

CORE COMPETENCIES

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| <ul style="list-style-type: none">Graphene anode development & surface modificationElectrode slurry prep, coating & calendaringIonic liquid & carbonate electrolyte formulationSEI engineering & interfacial stability optimizationCoin cell assembly: 2032 half-cell & full-cellBattery aging: capacity fade, CE, SoH, impedance | <ul style="list-style-type: none">Galvanostatic cycling, CV, EIS, dQ/dV (Arbin, Maccor)SOC/SOH monitoring & degradation analysisPre- and post-mortem electrode/electrolyte analysisSEM, XRD, FTIR, Raman spectroscopy, XPS, BETDesign of Experiments (DOE) & failure analysisPulsed Laser Deposition (PLD) & thin film fabrication | <ul style="list-style-type: none">N2 inert atmosphere glovebox operationCleanroom fabrication (NUANCE, NUFab at Northwestern)Moisture-sensitive materials handlingMulti-institutional R&D coordination & IP developmentTechnical documentation & patent prosecution supportVendor management & materials procurement |
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PROFESSIONAL EXPERIENCE

FONASE INC. (Battery Materials Startup)

Battery Materials Scientist & R&D Lead

Chicago, IL

Nov 2021 – Present

- Conceived and built a novel graphene-based Li-ion anode platform from zero as the sole technical contributor across the full development loop: active material synthesis, electrode slurry preparation, coating process development, calendaring, electrolyte formulation, separator selection, coin cell assembly, and electrochemical performance validation using Arbin and Maccor potentiostat systems.

- Developed two proprietary graphene electrode architectures via solution-based fabrication routes eliminating the need for industrial deposition equipment, designed for scalability to industrial manufacturing processes.
- Formulated and iterated ionic liquid electrolyte systems (Pyr14TFSI) with carbonate additives (FEC, ethylene sulfite) and lithium salts (LiTFSI, LiPF6), targeting SEI engineering outcomes including interfacial stability, first-cycle coulombic efficiency, and extended cycle life.
- Designed and executed electrochemical testing protocols for single-cell characterization across varied current densities and SOC windows, tracking CE, capacity fade, voltage efficiency, and SoH to drive iterative chemistry and process improvements.
- Conducted full materials characterization suite (SEM, XRD, FTIR, Raman spectroscopy, XPS, BET) to assess electrode morphology, graphitic structure, N/S doping levels, surface area, and surface chemistry across fabrication iterations.
- Performed pre- and post-mortem physicochemical analysis of electrolytes and electrodes to characterize SEI formation, interfacial degradation behavior, and capacity fade mechanisms across aging cycles.
- Named sole technical inventor on granted U.S. Patent No. 12,308,433; contributed to patent prosecution strategy and technical claim development with patent counsel.

External Research Collaborations

- **Northwestern University – Sponsored Research Collaboration (Evanston, IL):** Performed thin film deposition via Pulsed Laser Deposition (PLD) at Northwestern University Materials Research Center; conducted moisture-sensitive electrode fabrication and glovebox operation at GIANTFab (N₂ inert atmosphere); executed cleanroom fabrication at NUANCE and Nufab facilities (Aug 2022 – Dec 2024).
- **Illinois Institute of Technology (Chicago, IL):** Contracted half-cell assembly, electrochemical testing, and validation through the Thermal Processing Technology Center; managed technical requirements, data exchange, and iterative feedback across institutions.

EARLIER EXPERIENCE

Pennsylvania State University

State College, PA

Graduate Research Assistant

Jan – May 2021

- Conducted applied research in heterogeneous catalysis: assay development, metal extraction, and spectroscopic analysis (UV-Vis-NIR, ICP-OES). Non-thesis project: developed assay to differentiate metal-binding sites on single-atom ceria catalysts.

National Institute of Technology, Nagpur

Nagpur, India

Research Assistant

Jan – May 2018

- Developed and characterized nanoparticle adsorbents for wastewater treatment achieving 86.2% adsorption efficiency. Structural and surface analysis: XRD, FTIR, BET.

Industrial Trainee Rotations (2015–2017)

DRDO (Kanpur) – pilot plant design, batch reactor sizing, TEMA/ASME risk analysis for SiC foam production. | Dr. Reddy's Laboratories (Hyderabad) – Aspen Plus simulation, 15% throughput improvement. | HPCL (Visakhapatnam) – crude distillation and FCC unit inspection. | Visakhapatnam Steel Plant – coke oven operations, Kaizen/5S audits.

EDUCATION

Harrisburg University of Science and Technology

Harrisburg, PA

Master of Science, Project Management – PMP Track

Expected June 2026

Pennsylvania State University

State College, PA

Master of Science, Energy and Mineral Engineering

May 2021 | GPA: 3.75

Non-thesis: Developed assay to differentiate metal-binding sites on single-atom ceria catalysts

National Institute of Technology

Nagpur, India

Bachelor of Science, Chemical Engineering

May 2018 | GPA: 3.5

Thesis: Adsorption of impurities from effluent water using magnetic nanoparticles